Contact processes on ballistic Poisson particles: criticality, fronts, and when motion helps colonization

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Abstract

和文要旨. 実測恒星運動学を動機として、等速直線運動する Poisson 粒子系上の接触過程 $M(\rho, \nu, R, p, \tau, T_s)$(進入駆動・撃ちっ放し・対一回・再マーク許可) を導入し、その臨界構造を確立する。主結果: (i) 転送核のスペクトル条件 $r(K) < 1$ による絶滅定理 (定理 A)、(ii) この SIR 型分枝上界は再マーク許可の完全モデルには拡張されないこと (逆向きチャネルによる期待総子孫数 — 種を含むマーク事象総数 N — の分枝上界超過、4.4σ の数値確定)、(iii) 静的極限とボンド希釈ランダム幾何グラフの生存事象の同値 (定理 B)、(iv) 完全モデルの生存定理 (定理 C2 — セル・リレー構成、私有コイン結合、正規化橋、区間演算による機械認証)、(v) 「静的幾何が禁じる疎密度でも十分速い運動は入植を可能にする」(定理 D)、(vi) 前線の超線形/線形二分 (定理 C6a/C6b′)。さらに伝播規約への依存性 (進入駆動では運動が単調に助け、滞在時間要求型では最適速度が存在する) を数値的に確立する。検証は独立実装の数値反証装置・新規 AI インスタンスによる 11 サイクルの敵対的審査・幾何定数の機械認証の三層で行い、その全記録を開示する (付録 B)。

Abstract. Motivated by observed stellar kinematics, we introduce a contact process on particles moving ballistically in $\mathbb{R}^3$: a Poisson point process of intensity ρ whose points carry i.i.d. velocities drawn from ν, with entry-driven, fire-and-forget transmission at range R (success probability p, delay τ, each ordered pair attempts at most once ever), lifetimes $\mathrm{Exp}(T_s)$, and re-marking of dead points allowed. We establish the critical structure of this model. First, an extinction theorem: if the spectral radius of an explicit velocity transfer kernel is smaller than one, the SIR-type variant dies out almost surely (Theorem A), and the underlying moment computation is exact. Second, this branching-type upper bound *does not extend* to the full model with re-marking: a "reverse channel", by which a child re-marks its dead parent, produces a first-order excess of the expected total progeny (mark events including the seed) over the branching bound, certified numerically at 4.4σ (Section 4). Third, in the static limit the survival event coincides with percolation of a bond-diluted random geometric graph (Theorem B). Fourth, our main result: a survival theorem for the full model (Theorem C2), proved by a cell–relay renormalization whose geometric constants are certified by interval arithmetic, together with a private-coin coupling lemma, a disclosure-capsule identification, and a normalization bridge between accepted and disclosed candidate sets. Fifth, combining Theorems B and C2: at densities where static geometry forbids percolation for every p, sufficiently fast motion enables colonization (Theorem D). Finally we prove a superlinear/linear front dichotomy (Theorems C6a, C6b′),

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and establish numerically a *convention dichotomy*: under entry-driven transmission motion helps monotonically, while under dwell-time requirements an optimal speed emerges. All proofs were developed under a three-layer verification protocol (independent numerical falsification device, eleven cycles of adversarial line-by-line review by fresh AI instances, machine certification of all proof geometry), fully disclosed in Appendix B.

Contents

1 Introduction and main results

Consider a population of particles scattered in $\mathbb{R}^3$ as a Poisson point process, each moving forever along a straight line with a random velocity. A single marked particle starts an epidemic: whenever an unmarked particle *enters* the ball of radius R around a living marked particle, the mark is transmitted with probability p after a travel delay τ; marks die at rate $1/T_s$; dead particles can be re-marked. Does the epidemic survive forever? How fast does it spread? And — the question that motivated this work — *does the motion of the particles help or hinder*?

The model is a mathematical idealization of interstellar settlement fronts carried by probes among moving stars, in the convention of Carroll-Nellenback et al. [3] ("entry-driven, fire-and-forget": a settled system launches a probe when an unsettled system comes within range, and the probe travels once). The application of the resulting theory to observed stellar catalogs is deliberately separated into companion work; the present paper is self-contained probability theory.

Our model sits in a genuine gap between two literatures: epidemics on *diffusing* populations (Brownian motions with removal, Grimmett–Li [6]; spread of infection among random walks, Kesten–Sidoravicius [8, 9], Dauvergne–Sly [4]) and *static* continuum percolation (random geometric graphs, Penrose [16]; Poisson cylinders, Broman–Tykesson [2]). Ballistic (constant-velocity) motion is the kinematically correct regime for stars on ±few-Myr horizons, and it changes the mathematics: the space-time track of a particle is a straight cylinder, contacts between any ordered pair form a *single* time interval, and the environment de-correlates by transport rather than by mixing. A space-time percolation picture of this kind appears, non-rigorously, in delay-tolerant-network engineering [7]. Monotonicity of criticality in the velocity scale is open even in neighbouring models (jump-rate monotonicity for spread of infection; environment-speed monotonicity in evolving-environment contact processes, Linker–Remenik [14]), and is known to be direction-dependent across settings. Infections with recovery among moving particles are treated in [1]; percolated (bond-diluted) random geometric graphs, the static skeleton of our model, in [5, 11].

The model $M(\rho, \nu, R, p, \tau, T_s)$

At time 0, particles form a homogeneous Poisson process of intensity ρ on $\mathbb{R}^3$; particle i receives an i.i.d. velocity $v_i \sim \nu$ and moves as $x_i(t) = x_i + v_i t$ (the configuration is time-stationary). For an ordered pair (i, j) the contact set $\{t : |x_i(t) - x_j(t)| \leq R\}$ is a single interval $[\alpha_{ij}, \beta_{ij}]$. Transmission is *entry-driven and fire-and-forget*: if i is marked and alive during its contact window with j, then at time $a_{ij} = \max(\alpha_{ij}, m_i)$ (provided $a_{ij} < d_i$ and $a_{ij} \leq \beta_{ij}$) the pair attempts transmission *once and only once ever* (pair-once is an absolute convention; re-marking never re-opens a pair), succeeding with probability p; on success j becomes marked at time $a_{ij} + \tau$. Mark lifetimes are $\mathrm{Exp}(T_s)$; a particle whose mark has died may be marked again by a fresh pair (*re-marking*). The seed is one marked particle near the origin at $t = 0$. *Survival* means the set of living marked particles is non-empty at arbitrarily large times.

Two dimensionless quantities organize the phase diagram: with $\lambda := \rho\pi R^2\, \mathbb{E}|v - v'|$ (entry rate; $v, v' \sim \nu$ independent) and $\deg := \frac{4\pi}{3}\rho R^3$,

$$m := p\,(\deg + \lambda T_s), \qquad K_{\mathrm{mix}} := \frac{w_1 T_s}{R} \quad \text{(mixing number of a speed shell } w_1\text{)}.$$

Main results[1]

Write $r(K)$ for the spectral radius on $L^2(\nu)$ of the transfer kernel

$$K(v, v') \;=\; p\rho\Big(\tfrac{4\pi}{3}R^3 + \pi R^2\,|v - v'|\,T_s\Big).$$

Theorem A (extinction, SIR variant). *Assume $\mathbb{E}|v|^2 < \infty$. If $r(K) < 1$, the no-re-marking (SIR) variant dies out almost surely. The underlying first-moment (Mecke) computation of the expected number of length-k transmission chains is exact, so the spectral condition is sharp at the level of first moments.*

Claim (the SIR bound does not extend — numerically certified, experiment E8c; Section 4). *There are parameter regions with scalar mean $\bar{m} < 1$ in which the full model satisfies $\mathbb{E}[N] > 1/(1 - \bar{m})$, where N is the total progeny: the total number of mark events, including the seed. Consequently, the SIR-type (branching-process) threshold description does not extend to the full model with re-marking: the excess is produced by a first-order "reverse channel" (a child re-marks its dead parent through the unused reverse pair). Whether the extinction threshold itself separates — i.e., whether the full model can survive somewhere inside $r(K) < 1$ — is open (Section 10).*

Theorem B (static limit). *Let $\nu = \delta_0$. Then the survival event of $M(\rho, \delta_0, R, p, \tau, T_s)$ coincides (up to null sets) with the event that the origin's cluster is infinite in the bond-diluted random geometric graph $G(\rho, R; p)$, for every T_s and τ. In particular survival does not depend on T_s, τ in the static limit.*

[1] *Theorem labels retain the claim identifiers of the verification ledger (Appendix B), preserving traceability to the public record: some identifiers remain open conjectures (C4, Section 10), some were absorbed into other results during the sprint (C5 into Theorem B and the Section 8 synthesis), and C6 split into C6a and C6b′. Gaps in the numbering are deliberate.*

Theorem C2 (survival of the full model). *Let ν be isotropic with mass $q > 0$ on a speed shell $S = \{w : |w| \in [w_1, 2w_1]\}$, and with a density on S bounded by $\bar{\Lambda} \cdot q/\mathrm{vol}(S)$ with $\bar{\Lambda} \leq 4$ (relative density ratio; Maxwellian ν satisfies $\bar{\Lambda} \approx 2.4$ on its typical shell). Normalize $R = 1$ and assume the mixing condition $K_{\mathrm{mix}} = w_1 T_s \geq K_0 = 20$. There is a constant $C_0 = C_0(\tau/T_s) < \infty$ such that*

$$m_1 \;:=\; p\,\rho q\,\pi w_1 T_s \;>\; C_0 \qquad \Longrightarrow \qquad \textit{the full model survives with positive probability.}$$

For general $\bar{\Lambda} < \infty$ the same holds with $K_0 = K_0(\bar{\Lambda}) < \infty$ and $C_0 = C_0(\tau/T_s, K_0(\bar{\Lambda}))$.

Theorem D (motion beats static geometry). *Fix any $0 < \rho < \rho_c^{\mathrm{RGG}}(p{=}1)$, so that by Theorem B the static model dies out almost surely for every $p \leq 1$, T_s, τ. Then for every $p > 0$, $\tau \geq 0$, $T_s > 0$ there are isotropic velocity laws ν (scaled copies ν_s of any law satisfying the hypotheses of Theorem C2; the shell mass q and the ratio $\bar{\Lambda}$ are scale-invariant) such that the full model survives with positive probability. Under the entry-driven convention, sufficiently fast motion enables colonization at densities where static geometry forbids it.*

Theorem C6a (superlinear front). *If $T_s = \infty$ and ν is isotropic with unbounded support, then the front radius satisfies $R(t)/t \to \infty$ almost surely.*

Theorem C6b′ (linear bound for bounded speeds). *If $|v| \leq w_{\max}$ a.s. and $\tau > 0$, then for all $t \geq 0$, $R(t) \leq w_{\max} t + R\lfloor t/\tau \rfloor + |x_0|$, where x_0 is the seed position. In particular the front is at most linear, and $\tau > 0$ is precisely what guarantees well-foundedness of ancestries (Section 8).*

Numerically we complement the dichotomy: unbounded (Gaussian) velocity laws produce an accelerating front (fitted growth exponent 1.34 ± 0.14) while bounded laws are linear (0.98 ± 0.01); and we establish the *convention dichotomy*: under the entry-driven convention survival probability is monotone increasing in the velocity scale, while under a dwell-time-requirement convention an optimal speed $\sigma^* \approx 0.4$ emerges (Section 8).

Method of verification

Every claim in this paper was developed under a claim-driven verification protocol: an independent simulation device for numerical falsification, adversarial line-by-line review of each completed proof by fresh AI instances with no shared context (eleven review cycles for Theorem C2), and machine certification, by interval arithmetic, of every geometric constant in the survival proof. Section 9 states the exact boundary of what this internal verification does and does not establish; Appendix B is the complete method record, including the error ledger.

2 Model and conventions

We collect the conventions used throughout; they match the simulation device of Appendix A exactly.

Dynamics. Poisson(ρ) initial positions on $\mathbb{R}^3$; i.i.d. velocities $v_i \sim \nu$; ballistic motion $x_i(t) = x_i + v_i t$. At every time t the positions again form Poisson(ρ) (time stationarity). Since relative velocities are constant, the contact set of an ordered pair is a single interval; the entry time $a_{ij} = \max(\alpha_{ij}, m_i)$ lies in the contact window whenever an attempt occurs, so the transmission jump in space is at most R.

Transmission conventions. *Entry-driven*: attempts are triggered by entry events (or by the marking of i if j is already inside range). *Fire-and-forget*: no dwell requirement — the attempt is resolved instantly by a fair p-coin, the effect arriving after delay τ. *Pair-once*: each ordered pair (i, j) attempts at most once in the entire history; this is an absolute convention, unaffected by re-marking. *Re-marking*: a dead particle is eligible for fresh marks through fresh pairs. Lifetimes are $\mathrm{Exp}(T_s)$, independent of everything else; marks never kill and never displace — the process is purely additive on marks.

Scaling. $R = 1$ may be assumed (space-time scaling); m, m_1, K_{mix}, τ/T_s, and the ratio $\bar{\Lambda}$ are scale-invariant. In particular the pushforward $\nu_s = \nu^0(\cdot/s)$ of a shell law keeps q and $\bar{\Lambda}$ fixed while scaling K_{mix} and m_1 linearly in s — the scale-invariance lemma behind Theorem D.

3 Extinction: Theorem A

The proof (Appendix C.1) is a first-moment argument made exact. For a time-ordered chain of transmissions $0 \to i_1 \to \cdots \to i_k$, conditioning on velocities and applying the multivariate Mecke equation [10] for the Poisson process of (position, velocity) marks reduces the expected number of such chains to a k-fold operator power. Two technical points make the computation exact rather than merely an upper bound.

First, the *capsule volume lemma*: for a source of velocity v and window length T, the set of initial conditions (x_0, v') whose trajectory enters the moving ball within the window is a capsule of exact volume $\frac{4\pi}{3}R^3 + \pi R^2 |v - v'| T$; no overcounting occurs because entry times are unique (single contact interval).

Second, the *deterministic attempt-time recursion*: along a chain, the j-th attempt time satisfies $t_j = \max(\inf T_j,\ t_{j-1} + \tau)$, a deterministic function of the geometry, which quantifies the windows without conditioning on the past — this removes the window-quantification gap found in review (Appendix B, cycle 1).

Chains do *not* factor into scalar means: the velocity of the relay particle is shared between consecutive links, so the correct object is the kernel K acting on $L^2(\nu)$, not the scalar $\bar{m} = \int K \,\mathrm{d}\nu \,\mathrm{d}\nu$. If $r(K) < 1$, the expected total progeny is finite and the SIR variant dies out a.s.

4 The SIR-type bound does not extend to re-marking

It is tempting to expect Theorem A to hold verbatim for the full model with $\bar{m}$-type conditions. It does not.

Claim (reverse channel — numerically certified, experiment E8c). *There exist parameters with $\bar{m} < 1$ (e.g. near-static pairs, $p = 1$, $\tau = 10\,T_s$, $\deg = 0.1$) for which the full model satisfies $\mathbb{E}[N] > 1/(1 - \bar{m})$, where N is the total progeny (mark events including the seed).*

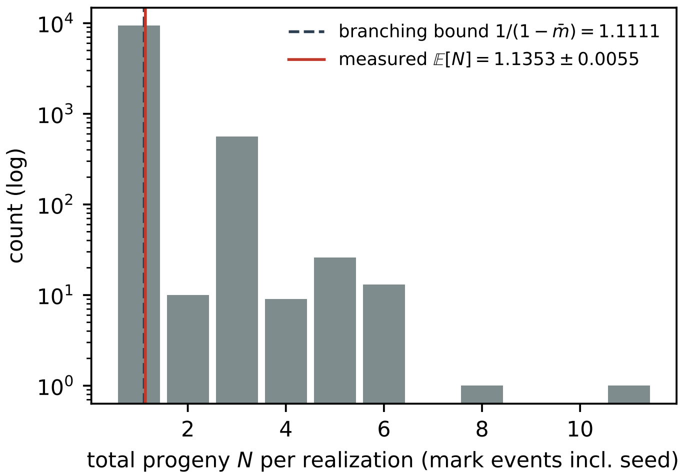

Figure 1: Reverse channel (experiment E8c): histogram of the total progeny N (mark events including the seed) over 10^4 realizations in the counterexample regime. The measured mean exceeds the branching bound by 4.4σ. Reproduce: `python3 src/wake_r/paper_figs.py` (data: `data/phase_r/E8c_counterexample.json`).

The mechanism is a *first-order self-influence* that Mecke chain counting cannot see: a child j, marked via (i, j), can re-mark its *dead* parent i through the never-used reverse pair (j, i) — the probability per infection is of order $p(1 - e^{-2\tau/T_s})$ and the channel lives in the corner of slow relative velocities and $\tau \gtrsim T_s$. The device certifies the violation: over 10^4 independent realizations in the counterexample regime, $\mathbb{E}[N] = 1.1353 \pm 0.0055$ against the branching bound $1/(1 - \bar{m}) = 1.1111$, a 4.4σ excess (Figure 1). We emphasize the exact strength of the statement: this is a certified violation of the *expectation bound*, hence of the SIR-type threshold *description*; it does not by itself prove survival anywhere inside $r(K) < 1$, and the separation of the extinction thresholds themselves remains open (Section 10).

5 Static limit: Theorem B

Let $\nu = \delta_0$. Then all contact windows are $[0, \infty)$ or empty, every attempt happens at the instant its source is marked, and delays only shift the marking times. The set of particles ever marked is exactly the cluster of the seed in the graph whose vertices are the Poisson points and whose edges are pairs at distance $\leq R$ retained independently with probability p — the bond-diluted random geometric graph $G(\rho, R; p)$ [16, 15, 5]. The *survival event* therefore coincides a.s. with the percolation event of the seed's cluster, for every T_s, τ; lifetimes only govern how long the (predetermined) cluster stays lit. A double numerical verification (dynamic device vs. independent union-find implementation) is recorded in Appendix A.

6 Survival: Theorem C2

This is the technical core of the paper. Full proofs are in Appendix C.3; here we present the architecture and the five load-bearing lemmas. Constants below refer to the certified configuration $K_0 = 20$, $T_a = T_s/32$, $T_g = 64(\tau + T_s)$.

6.1 Cell–relay construction

By the pigeonhole principle, one of the 200 equal-width sub-shells of $[w_1, 2w_1]$, written I^*, carries mass $\geq q/200$; let $\bar{w}$ be its midpoint. The *aiming constant* $\bar{u}_x := \bar{w}/\sqrt{17/16}$ makes the ideal required velocity $(\bar{u}_x, \pm\bar{u}_x/4, 0)$ have norm exactly $\bar{w}$. Sites (n, k), $n + k$ even, of a two-dimensional oriented lattice are realized as spatial boxes of side $2a$, $a = \bar{u}_x T_g/32$, with y-tracks spaced $8a$ apart and a single z-track; anchoring is *hybrid* (y, z global, x relative to the parent, targets rounded to the lattice $(a/8)\mathbb{Z}^3$). A site is *open* if the relay delivered at least N_h marked, living particles with velocities in a small cover box into the box core (half-width $a/2$) at time t_k, at least one of which has residual lifetime $\geq T_g$ (*coverage clause* — this keeps the marked set non-empty between epochs). From a good parent P^*, the relay uses entries into its radius-1 capsule during the *entry window* $[t_k, t_k + T_a]$ only.

6.2 The five lemmas

1. **Geometry (machine-certified).** Over all perturbations (parent position in core + rounding, parent velocity in the $\pm 0.05\bar{w}$ cover, entry offset $\leq R$, window sweep $s \in [0, T_a]$, branch signs), the landing window $w_c \oplus [-h_0, h_0]^3$, $h_0 = \frac{a/2}{T_g - s}$, retains a margin $h_0 - \text{dev} - I_h - \text{lip} \geq s^* = 0.0048\,\bar{w}$ (at $\tau = 0$; larger for $\tau > 0$), stays inside the velocity cone and inside the parent cover. Certification is by interval arithmetic with exact norm bounds and a Lipschitz penalty on an s-grid (Appendix A).

2. **Erosion patch.** With $\theta^* = s^*/(1.05\bar{w})$, the full angular patch of radius θ^* around the aim direction, times the whole radial band I^*, lands in the window: by isotropy, $\hat{\nu}(\text{accepted}) \geq \frac{1}{200} \cdot \frac{1 - \cos\theta^*}{2}$.

3. **Disclosure capsule identification.** The construction reads the randomness of *window entrants only*: the single-contact-interval property makes a window entry a *first* contact, so pair-once guarantees the attempt fires inside the window; entries outside windows never enter the construction's σ-algebra. Hence the deduction tubes below have segment length exactly $|v - v_P| T_a$ — disclosure geometry, not dynamical exposure.

4. **Normalization bridge (machine-certified).** Candidates contaminated by past sites are those in the sets $\{v : \tilde{\Delta} - v\, jT_g \in \mathcal{K}\}$, $\mathcal{K} = B(0, 2R) \oplus \text{seg} \oplus \text{seg}$; the union over the window is contained in a *box-hull* $B(0, 2R) \oplus H_1 \oplus (-H_2)$ whose exact Steiner volume is $\prod c_i + 2r \sum_{i<j} c_i c_j + \pi r^2 \sum c_i + \frac{4\pi}{3} r^3$ ($r = 2R$). With the site-count bound $n_j \leq \frac{\sqrt{3}}{8} j + 1 + \operatorname{diam}\mathcal{K}/(8a)$ (single z-track; gap j fixes the generation) and the density hypothesis, the disclosed mass satisfies $\bar{\Lambda}\,\hat{\nu}(\text{bad}) \leq \frac{1}{2}\hat{\nu}(\text{accepted})$ for $K_0 = 20$, $T_a = T_s/32$, $\bar{\Lambda} \leq 4$: at $\tau = 0$ (worst case) the certified values are $\hat{\nu}(\text{bad})|_{\bar{\Lambda}=1} = 1.69 \times 10^{-9}$ versus $\hat{\nu}(\text{accepted}) = 2.64 \times 10^{-8}$, i.e. the bridge closes up to $\bar{\Lambda}_{\max} = 7.8$, about twice the assumed bound.

5. **Private-coin coupling (Lemma X).** Conditional on the entire disclosure, the failure of a pair-fresh candidate to be marked-and-alive at the epoch end requires the shortfall of a *single private exponential coin* of that candidate; the conditioning reads lifetimes only through indicator paths $(\mathbf{1}\{L > s\})_{s \leq a}$, under which the relevant residuals remain jointly independent $\text{Exp}(T_s)$; and post-arrival interference can only *rescue* (re-marking never

harms — additivity). Hence the number of delivered children stochastically dominates a Binomial with success probability $\beta = e^{-T_g/T_s}$ over a Poisson number of candidates with mean $\mu \geq c_4(\tau/T_s)\, m_1/32$, surviving the bridge deduction at least in half.

6.3 Assembly

Chernoff bounds give $\mathbb{P}(\text{site open} \mid \text{all disclosure}) \geq 1 - \varepsilon(m_1)$ with $\varepsilon \to 0$ as $m_1 \to \infty$; by the tower property this holds sequentially in any breadth-first order, so the open sites dominate an i.i.d. Bernoulli$(1-\varepsilon)$ field on the oriented lattice (direct sequential conditioning; cf. the domination theory of [13], which our argument does not need). The comparison lattice is precisely $\mathcal{L} = \{(n,k) \in \mathbb{Z} \times \mathbb{Z}_{\geq 0} :\ n+k \text{ even}\}$ with adjacency $(n,k) \to (n \pm 1, k+1)$ — *the same adjacency structure* for which Liggett's bound $p_c^{\text{site}} \leq 3/4$ is proved [12]; we verified the identification of the two lattices explicitly (each site has exactly the two children $(n \pm 1, k+1)$ and the two parents $(n \mp 1, k-1)$, and openness is a site property). For $m_1 > C_0$ we have $1 - \varepsilon > 3/4$, hence supercritical oriented site percolation, an infinite open cluster with positive probability, and — through the coverage clause — a marked living particle at every time. Ignition from the arbitrary-velocity seed is a positive-probability bootstrap. □

Remark 1. *No upper bound on m_1 is used anywhere: the coverage clause, the Chernoff step and the observation truncation $M = \lceil 2\mu \rceil$ all improve or are unaffected as m_1 grows. Intermediate τ are covered by monotonicity of both sides of the bridge. The pure-isotropy case (no density bound) is left as a conjecture (Section 10).*

7 Motion beats static geometry: Theorem D

Fix $0 < \rho < \rho_c^{\text{RGG}}(p{=}1)$. By Theorem B and the standard monotone coupling in p, the static model dies for all $p \leq 1, T_s, \tau$. Now fix any $p > 0, \tau \geq 0, T_s > 0$ and any base law ν^0 satisfying the hypotheses of Theorem C2. Scaling $w \mapsto sw$ preserves the shell mass q and the relative density ratio $\bar{\Lambda}$ (the density scales by s^{-3} and the shell volume by s^3), while $K_{\text{mix}} = s\, w_1^0 T_s$ and $m_1 = p\rho q\pi\, s\, w_1^0 T_s$ grow linearly in s with τ/T_s fixed; for s large both hypotheses of Theorem C2 hold, and the full model survives with positive probability. □

8 Fronts and the convention dichotomy

8.1 Superlinear and linear fronts

Theorem C6a (Appendix C.4) shows $R(t)/t \to \infty$ a.s. for $T_s = \infty$ and isotropic unbounded ν: the seed alone recruits carriers of ever-larger speeds, an infinite-system version of the fastest-of-N heuristic used in [3]. Conversely, Theorem C6b′: for $|v| \leq w_{\max}$ and $\tau > 0$,

$$R(t) \ \leq\ w_{\max}\, t \ +\ R\left\lfloor \tfrac{t}{\tau} \right\rfloor \ +\ |x_0|.$$

The proof is a strict accounting: along any ancestry of mark events, marking times increase by at least τ per hop — which simultaneously proves the ancestry is *well-founded* (no infinite descent; this, not locality per se, is the true role of $\tau > 0$) — and travel time telescopes exactly across $[0,t]$, each hop contributing a spatial jump $\leq R$ at its attempt time. The companion

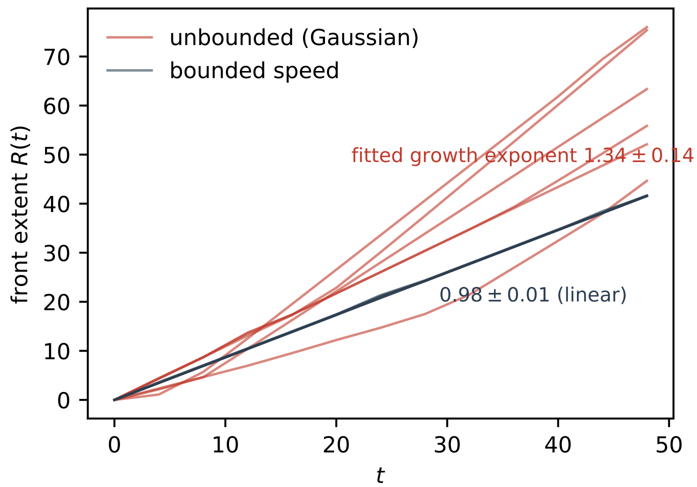

Figure 2: Front dichotomy (experiment E5): front extent $R(t)$ for unbounded (Gaussian) versus bounded velocity laws, six runs each. Reproduce: `python3 src/wake_r/paper_figs.py` (data: `data/phase_r/E5_front.json`).

pathology: at $\tau = 0$, if the open cluster of the seed in the p-bond-diluted instantaneous geometric graph is infinite, the whole cluster is marked at time 0^+ and $R(0^+) = \infty$ can occur.

Numerically the dichotomy is sharp (Figure 2): Gaussian velocity laws accelerate (fitted exponent 1.34 ± 0.14), bounded laws are linear (0.98 ± 0.01).

8.2 The convention dichotomy

Does motion help colonization? Theorem D answers "yes, decisively" for the entry-driven convention, and experiment E4c confirms strict monotonicity of survival probability in the velocity scale ($0.73 \to 0.91$ over $\sigma \in [0, 0.16]$ at fixed ρ, p). But replace fire-and-forget with a *dwell-time requirement* (transmission requires residence within range for a fixed time) and the answer inverts: experiment E7 exhibits a non-monotone survival curve with an optimal speed $\sigma^* \approx 0.4$ (survival fraction $0.06 \to 0.31 \to 0.00$), consistent with the optimal-velocity phenomenology observed in mobile-agent epidemic models [17, 18]. *The answer to "does motion help" is a function of the transmission convention*, exhibited here within a single model family (Figure 3). A related trio of non-monotonicities (in particle number, in re-marking, in timing — all SIR-type interference effects) is recorded in the method archive.

9 Verification: method and boundary

Every mathematical claim above was developed under a claim-driven protocol with three layers: (i) an independent simulation device (analytic closest-approach entry detection, time-step independent) used to attempt numerical falsification of every stated claim; (ii) adversarial, line-by-line review of each completed proof by fresh AI instances sharing no context with the author system — Theorem C2 passed on the eleventh cycle, and the complete cycle ledger, including more than ten caught errors (five geometric-arithmetic, one non-convexity, one tube-

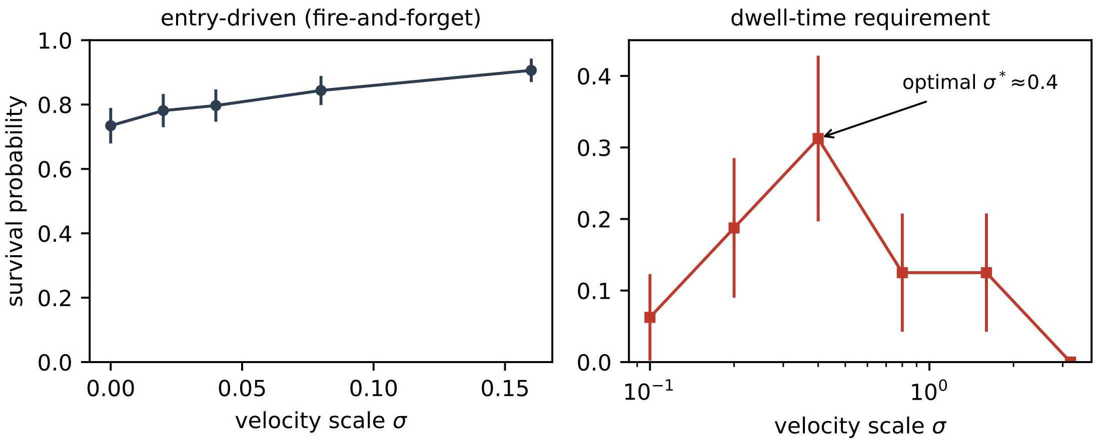

Figure 3: Convention dichotomy (experiments E4c, E7): survival probability versus velocity scale σ. Left: entry-driven fire-and-forget — monotone increasing. Right: dwell-time requirement — optimal speed $\sigma^* \approx 0.4$. Error bars are binomial standard errors. Reproduce: `python3 src/wake_r/paper_figs.py` (data: `data/phase_r/{E4c_dip,E7_dwell}.json`).

direction, one normalization-bridge gap, one exposure-length gap, one over-general lemma), is disclosed in Appendix B; (iii) machine certification, by interval arithmetic, of every geometric constant of the survival proof, including the normalization bridge (Appendix A). All bibliography entries carry an arXiv identifier or DOI and were mechanically resolved by the verification script of Appendix A; the resolution log is included in the bundled repository archive (`wake-repo-v1.tar.gz`).

Boundary. This internal verification — independent AI review instances plus machine certification — *is not a substitute for human peer review.* The role of AI systems in this work, and the exact experimental conditions under which the mathematics was produced (including the empty steering-intervention log), are disclosed in Appendix B, which also serves as the AI-use disclosure in the sense of current arXiv policy. The author takes full responsibility for all content.

10 Open problems

1. **Pure isotropy.** Remove the bounded-density hypothesis ($\bar{\Lambda}$) from Theorem C2.

2. **Threshold separation.** Does the full model survive anywhere inside $r(K) < 1$? (The reverse channel breaks the branching *bound*; whether it moves the *threshold* is open.)

3. **Repaired full-model extinction.** An extinction condition for the full model incorporating the reverse channel into the transfer kernel; without additional assumptions this appears hard (Appendix B, cycle 4).

4. **Velocity monotonicity (C4).** Pathwise coupling monotonicity of survival in the velocity scale for the entry-driven model; open, in line with neighbouring open problems [8, 14]. The non-monotonicity trio shows naive couplings must fail.

5. **Front shape theorem.** Existence of the limiting front shape and speed for bounded ν; explicit constants for C_0 and critical exponents.

Acknowledgements

The mathematical exploration, proofs, and their revisions were carried out by AI systems (Claude Code, model Claude Fable 5) under the verification protocol of Section 9, with the human author acting as arbiter (go/stop/harvest rulings and encouragement only; the steering-intervention log is empty). See Appendix B for the complete disclosure.

A Machine certification bundle

A.1 Contents

The complete repository snapshot is bundled with this Zenodo record as `wake-repo-v1.tar.gz` (all version-tracked files at the release commit; regenerate with `git archive`). It contains: `src/wake_r/geo_verify.py` (certifying verifier, version 6; deterministic, standard library only), `docs/phase-r/certification-log-v6.txt` (full scan log), `src/wake_r/device.py` (falsification device), `src/wake_r/paper_figs.py` (figure generation), `data/phase_r/*.json` (experiment archive E1–E8c), and `scripts/verify_refs.py` with its resolution log (bibliography verification). Reproduction: `python3 src/wake_r/geo_verify.py`.

A.2 Design

All proof geometry is certified by interval arithmetic: every perturbation (parent position within core plus rounding, parent velocity within the $\pm 0.05\bar{w}$ cover, entry offset $\leq R$, window sweep, branch signs, target rounding) is propagated as component intervals; Euclidean norms are bounded by the exact closed forms $\min \|x\|^2 = \sum_i \operatorname{dist}(0, [l_i, h_i])^2$ and $\max \|x\|^2 = \sum_i \max(l_i^2, h_i^2)$; the window sweep uses a 4096-cell grid with a Lipschitz penalty $|\partial_s w_{\mathrm{req}}| \leq (\sup |v_p| + \sup |w_{\mathrm{req}}|)/(T_g - T_a)$ and the conservative left endpoint for h_0. The bridge certification computes the acceptance mass $\frac{1}{200} \cdot \frac{1-\cos\theta^*}{2}$ and the disclosed mass via the exact box–ball Steiner volume and the site-count series $\frac{\sqrt{3}}{8}\zeta(2) + (1 + \operatorname{diam}\mathcal{K}/8a)\zeta(3)$, with the worst shell endpoint taken separately on each side. All approximations are monotone in the conservative direction; the box-hull correctness is the direct inclusion $\bigcup_v \mathcal{K}(v) \subseteq B(0, 2R) \oplus H_1 \oplus (-H_2)$.

A.3 Certified table (excerpt)

K_0	τ/T_s	slack/$\bar{w}$	bad ($\bar{\Lambda}$=1)	accepted	$\bar{\Lambda}_{\max}$
20	0	+0.00483	1.688×10^{-9}	2.640×10^{-8}	**7.82**
20	1	+0.00536	2.100×10^{-10}	3.253×10^{-8}	77.47
20	3	+0.00562	2.618×10^{-11}	3.583×10^{-8}	684.33
24	0	+0.00499	1.091×10^{-9}	2.821×10^{-8}	12.93

Verdict: $K_0 = 20$, $\bar{\Lambda} \leq 4$, $T_a = T_s/32$ — PASS (bridge and geometry), with margin factor ≈ 1.95 over the assumed $\bar{\Lambda} \leq 4$. For comparison, the pre-repair configuration $K_0 = 5$

yields $\bar{\Lambda}_{\max} = 0.03$: the bridge genuinely fails there, exactly as found in review cycle 10 (Appendix B). Independent reproductions of these numbers by fresh review instances are recorded in Appendix B.

B Method record (self-contained)

This appendix is a self-contained disclosure document; it can be read independently of the main text. It records the experimental conditions under which the mathematics of this paper was produced: the protocol separating encouragement from mathematical steering, the (empty) intervention log, the complete review-cycle ledger with all caught errors, and the evolution of the machine-verification layer.

B.1 Design

The executor (an AI system: Claude Code, model Claude Fable 5) held full authority over mathematical exploration — proof strategies, experiment design, retraction decisions. The human author acted solely as arbiter (go/stop/harvest rulings) and encourager; an independent AI chat instance relayed verification verdicts only. Claim-driven verification: on every declared claim, (i) numerical falsification against an independently implemented simulation device, (ii) literature check; on every completed proof, line-by-line adversarial review by a *fresh* AI instance sharing no conversational context, one new instance per cycle.

B.2 Steering protocol and intervention log

Permitted interactions: encouragement, confidence, generic advice ("stepping away is allowed"), and the go/stop/harvest ruling. Forbidden: mathematical hints, literature pointers, evaluations of proof directions. Any steering intervention was to be logged as a change of experimental conditions. **The intervention log is empty**: from sprint start (2026-08-15) to close (2026-08-16), no forbidden intervention occurred; all human inputs were start permissions, encouragement, and rulings.

B.3 Review-cycle ledger

#	target	verdict	errors caught / outcome
self	Thm. A v1	self-rejected	chain factors share velocities; scalar $\bar{m}^k$ invalid $\to$ spectral $r(K)$
1	Thm. A/C6a	accepted after fix	window quantification gap $\to$ deterministic attempt-time recursion
2	C2 v1	rejected	empty landing set; co-moving particles break finite range; no thinning monotonicity
3	C2 v2	core isolated	D-separation impossible; general memoryless lemma false (negative-information)
4	A-full	**retracted (false)**	reverse-channel counterexample $\to$ device-certified 4.4σ; became Section 4
5–8	C2 v3–v6	partial accepts	five geometric-arithmetic errors; $\|w\|$ non-convexity $\to$ interval arithmetic
9	C2 v7	remanded	tube direction (relative velocity); honest Minkowski sums fail $\to$ window shortening
10	C2 v8	**incomplete**	normalization bridge missing (real: 33–440× short); exposure-length gap
11	C2 v9	**accepted**	over-general sector lemma $\to$ replaced by box-hull Steiner; numbers reproduced
spot	box-hull, Thm. D	accepted	scale-invariance lemma made explicit; false "superset" justification removed
C6b	Prop. C6b	accepted (minor)	tightened to $\lfloor t/\tau \rfloor$; well-foundedness made explicit

Tally: more than ten errors caught; two became discoveries (reverse channel, non-monotonicity trio), one became a strengthening (window shortening); one theorem retracted and re-cast as a correct impossibility statement (Section 4, with the claim strength corrected in final editing: what is proved is that the branching-type bound does not extend, not that the thresholds separate).

B.4 Machine-verification evolution

Monte-Carlo minimum (rejected in review: not a worst-case proof) $\to$ corner enumeration with Lipschitz penalties (rejected: anti-conservative on concave directions of the norm) $\to$ interval arithmetic with exact norm bounds and bridge certification (final; also caught two design errors before they reached review). Operational conclusion: *every geometric constant of a proof of this kind should be machine-certified*; hand-computed geometry failed five times in review.

B.5 Disclosure statement

The mathematical content of this paper (formulations, proofs, retraction decisions included) was explored by an AI executor; verification used the three layers above. The human author's role was direction-level arbitration and encouragement; the steering log is empty. This internal verification is not a substitute for human peer review; the reader and referee should treat this record, including the error ledger, as the disclosure of the verification boundary.

C Proof details

This appendix contains the complete proofs of Theorems A, B, C2, C6a, C6b′ and D, in their accepted forms (the review record is Appendix B). Notation is that of Section 2. All certified numerical values (margins, disclosed and accepted masses, $\bar{\Lambda}_{\max}$) are cited from the certified table of Appendix A.3 — the single-source principle; the certifying program is described in Appendix A.

C.1 Theorem A: extinction of the SIR variant

Throughout this subsection the SIR (no-re-marking) variant is in force: a particle carries a mark at most once in its history; all other conventions are those of Section 2. We assume $\mathbb{E}|v|^2 < \infty$ and prove: if $r(K) < 1$, the expected total number of mark events is finite and the SIR variant dies out almost surely; moreover the underlying chain count is an equality, so the spectral condition is sharp at the level of first moments. As in Section 3, chains do not factor into scalar means — the relay velocity is shared by consecutive links — so the correct object is the kernel K on $L^2(\nu)$.

(a) Generations and the chain tree. Add the seed as an extra particle at the origin with velocity $v_0 \sim \nu$ (Palm version); it is marked at time $m_0 = 0$, with mark lifetime $L_0 \sim \mathrm{Exp}(T_s)$. The marked set is constructed by generation induction: generation 0 is the seed; a particle belongs to generation at most $k+1$ if it receives a successful attempt from a particle of generation at most k. Every mark event other than the seed has a unique parent (the source of the successful attempt that created it; in the SIR variant mark events are identified with particles), and by the inductive construction every marked particle is joined to the seed by a finite parent chain: the parent relation is a well-founded tree and depth is well defined. A depth-k mark determines the time-ordered chain $(i_0 = \mathrm{seed}, i_1, \ldots, i_k)$ of *distinct* particles along its ancestry.

(b) The deterministic attempt-time recursion. For an ordered pair write $T_{ij} = [\alpha_{ij}, \beta_{ij}]$ for its contact interval (a single interval, by ballistic motion). Along a chain abbreviate $T_j := T_{i_{j-1}i_j}$ and define recursively

$$m_0 := 0, \qquad t_j := \max(\inf T_j,\ m_{j-1}), \qquad m_j := t_j + \tau \quad (1 \le j \le k).$$

Call the chain *open* if for every j the pair coin of (i_{j-1}, i_j) succeeds and

$$T_j \cap [m_{j-1},\, m_{j-1} + L_{j-1}) \ \neq\ \emptyset,$$

where L_{j-1} is the lifetime of i_{j-1}'s mark. If the chain is realized by actual mark events, the actual attempt times are exactly $a_j = \max(\alpha_j, m_{j-1}) = t_j$, and the attempt requires the parent to be alive ($t_j < m_{j-1} + L_{j-1}$) and in contact ($t_j \in T_j$); hence realized chains are open — the open-chain event is a necessary condition (a relaxation), and the number of depth-k marks is at most the number N_k of open chains of length k. Two features make the event usable. First, t_j is a *deterministic* function of the contact geometry, so the windows are quantified without conditioning on the past dynamics; this removes the window-quantification gap found in review — in the first version, nested existential witnesses inflated the window by the contact duration, which cannot be absorbed into the kernel (Appendix B, cycle 1).

Second, the window $[m_{j-1}, m_{j-1} + L_{j-1})$ has length exactly L_{j-1}, and τ enters only through the translation $m_j = t_j + \tau$ of the windows; by the translation invariance of the capsule volume below, τ contributes nothing to the moment computation — *this is why the kernel K contains no τ.*

(c) The capsule volume lemma.

Lemma 1 (capsule volume)**.** *Let a source move as $x(t) = x + vt$ and fix a time window $[s, s+\ell)$, $\ell > 0$. For $w \in \mathbb{R}^3$, the set of initial positions $y \in \mathbb{R}^3$ whose trajectory $y + wt$ satisfies $\{t : |y + wt - x(t)| \le R\} \cap [s, s+\ell) \neq \emptyset$ is a capsule of volume exactly*

$$\tfrac{4\pi}{3} R^3 + \pi R^2 \, |v - w| \, \ell,$$

independent of the window position s and of x.

Proof. In relative coordinates $z = y - x$, $u = w - v$ the condition reads $z + ut \in B(0, R)$ for some $t \in [s, s+\ell)$, i.e. $z \in B(0,R) \oplus \{-ut : t \in [s, s+\ell)\}$: a translate of the ball swept along a segment of length $|u|\ell$ — a spherocylinder of volume $\frac{4\pi}{3}R^3 + \pi R^2 |u| \ell$ (the half-open end is Lebesgue-null). □

Because the contact set of every ordered pair is a single interval, the entry time into any window is unique; hence in the chain count below the integrated indicator is exactly capsule membership — no configuration is counted with multiplicity, and the volume above is exact, not an upper bound.

(d) The Mecke computation. The particles form a Poisson process Ξ on phase space $\mathbb{R}^3 \times \mathbb{R}^3$ with intensity $\rho \, \mathrm{d}x \otimes \nu(\,\mathrm{d}v)$. The number of open chains of length k is

$$N_k = \sum_{\substack{(y_1,w_1),\dots,(y_k,w_k) \in \Xi \\ \text{distinct}}} \mathbf{1}\big\{(\text{seed}, (y_1, w_1), \dots, (y_k, w_k)) \text{ open}\big\}.$$

The open-chain event depends only on the k tuple points, the seed, the lifetimes $L_0, \dots, L_{k-1}$ and the k pair coins — not on the remaining configuration — and the k ordered pairs along a chain of distinct points are pairwise distinct, so their coins are i.i.d. and contribute the factor p^k. By the multivariate Mecke equation for Poisson processes [10],

$$\mathbb{E}[N_k] = \rho^k p^k \int_{(\mathbb{R}^3)^k} \int_{(\mathbb{R}^3)^k} \mathbb{E}\big[\mathbf{1}\{\text{window conditions hold}\}\big] \, \nu(\,\mathrm{d}w_1) \cdots \nu(\,\mathrm{d}w_k) \, \mathrm{d}y_1 \cdots \mathrm{d}y_k,$$

the inner expectation being over v_0 and the lifetimes. Integrate by Tonelli from the inside out. The window of step k is $[m_{k-1}, m_{k-1} + L_{k-1})$, and m_{k-1} depends only on $y_1, \dots, y_{k-1}$ and the velocities — not on y_k; by Lemma 1 the y_k-integral equals $\frac{4\pi}{3}R^3 + \pi R^2 |w_{k-1} - w_k| L_{k-1}$ regardless of the window position. The lifetime L_{k-1} occurs in step k only; averaging it ($\mathbb{E}L = T_s$) and multiplying by ρp yields exactly $K(w_{k-1}, w_k)$. Iterating for $j = k-1, \dots, 1$ (the seed's window is $[0, L_0)$, of length L_0) gives the *equality*

$$\mathbb{E}[N_k] \;=\; \int \nu(\,\mathrm{d}v_0) \, (K^k \mathbf{1})(v_0) \;=\; \langle \mathbf{1}, K^k \mathbf{1} \rangle_{L^2(\nu)}.$$

(e) The spectral step. The kernel $K(v,v') = p\rho\big(\frac{4\pi}{3}R^3 + \pi R^2|v-v'|T_s\big)$ is symmetric and nonnegative, and

$$\iint K(v,v')^2\,\nu(\mathrm{d}v)\nu(\mathrm{d}v') \;\le\; 2(p\rho)^2\Big[\big(\tfrac{4\pi}{3}R^3\big)^2 + (\pi R^2 T_s)^2\,\mathbb{E}|v-v'|^2\Big] < \infty$$

because $\mathbb{E}|v|^2 < \infty$. Hence K is a Hilbert–Schmidt integral operator on $L^2(\nu)$: bounded and self-adjoint, with $\|K\| = r(K)$; no positivity theory (Krein–Rutman) is needed. Since $\|\mathbf{1}\|_{L^2(\nu)} = 1$,

$$\mathbb{E}[\#\text{depth-}k\text{ marks}] \;\le\; \mathbb{E}[N_k] = \langle \mathbf{1}, K^k\mathbf{1}\rangle \;\le\; \|K\|^k = r(K)^k.$$

If $r(K) < 1$, the expected total number of mark events (seed included) is at most $\sum_{k\ge 0} r(K)^k = (1-r(K))^{-1} < \infty$. Almost surely there are finitely many mark events; each carries a finite lifetime, so $\max_i(m_i + L_i) < \infty$ and the set of living marked particles is empty from that time on: the SIR variant dies out almost surely. □

C.2 Theorem B: the static limit

Proof. Let $\nu = \delta_0$, so all particles are static and pair distances are constant in time. (1) *Windows*: the contact set of an ordered pair is $[0,\infty)$ if the distance is at most R, and empty otherwise. (2) *Instant attempts at marking times*: when a particle i becomes marked, at time m_i, every in-range pair (i,j) is already inside its window, so $a_{ij} = \max(\alpha_{ij}, m_i) = m_i$: the attempt fires at the instant of marking (the fresh mark is alive at its birth almost surely), is resolved instantly by the pair coin, and on success j is marked at $m_i+\tau$. (3) *Delays shift times only*: by induction, a particle marked through k hops is marked at time exactly $k\tau$; which particles are ever marked is decided by the coins and the geometry alone — T_s and τ never enter. (4) *Cluster identification*: the ever-marked set is therefore the forward-reachable set of the seed in the digraph on the Poisson points whose in-range arcs are passable according to their pair coins. In the breadth-first exploration of this reachable set, every in-range unordered pair with exactly one endpoint ever marked is consulted in one direction only (from the marked side), while pairs inside the marked set do not affect the cluster; declaring each in-range edge open according to the consulted coin (ties and never-consulted pairs by a fixed convention — such coins are no-ops for the marked set) realizes a bond-diluted random geometric graph $G(\rho,R;p)$ on the same probability space, in which the ever-marked set is exactly the cluster of the seed [16, 15]. If the cluster is finite, there are finitely many mark events, each of finite lifetime, and the living marked set is eventually empty forever. If the cluster is infinite, marks are created at every epoch $k\tau$ (at time 0^+ if $\tau=0$), and each freshly marked particle is alive at its marking instant, so living marked particles exist at arbitrarily large times, and lifetimes only govern how long the predetermined cluster stays lit. Hence the survival event coincides, up to null sets, with the percolation event {the seed's cluster in $G(\rho,R;p)$ is infinite}, for every T_s and τ. □

C.3 Theorem C2: survival of the full model

This subsection is the complete proof of Theorem C2 in its accepted form (review cycle 11 together with the accepted wording fixes; Appendix B). The architecture is summarized in Section 6; here every lemma is stated and proved.

Statement (Section 1). ν isotropic; shell $S = \{w : |w| \in [w_1, 2w_1]\}$ of mass $q := \nu(S) > 0$; on S, ν has a Lebesgue density bounded by $\bar{\Lambda} \cdot q/\mathrm{vol}(S)$ with $\bar{\Lambda} \le 4$; $R = 1$; mixing condition $K_{\mathrm{mix}} = w_1 T_s \ge K_0 = 20$. Then there is $C_0 = C_0(\tau/T_s) < \infty$ such that $m_1 = p\rho q\pi w_1 T_s > C_0$ implies survival with positive probability. *General-$\bar{\Lambda}$ clause*: for every $\bar{\Lambda} < \infty$ there is $K_0(\bar{\Lambda}) < \infty$ — the deduction side of the bridge below decays like K_0^{-2} while the acceptance side is monotone; the certified scan gives $\bar{\Lambda} \le 7.8$ at $K_0 = 20$ and $\bar{\Lambda} \le 12.9$ at $K_0 = 24$ (Appendix A.3) — and $C_0 = C_0(\tau/T_s, K_0(\bar{\Lambda}))$. The pure-isotropy case (no density bound) is open (Section 10).

C.3.1 Parameters, lattice, and the site event

Throughout $R = 1$; the certified configuration is the *entry window* $T_a := T_s/32$ and the *generation time* $T_g := 64(\tau + T_s)$, at $K_0 = 20$.

Sub-shell pigeonhole. Split $[w_1, 2w_1]$ into 200 sub-shells of equal width; one of them, written I^*, carries mass at least $q/200$; its half-width is $I_h \le \bar{w}/400$, where $\bar{w}$ denotes the midpoint of I^*.

Aiming constant. $\bar{u}_x := \bar{w}/\sqrt{17/16}$; the ideal required velocities $(\bar{u}_x, \pm\bar{u}_x/4, 0)$ have norm exactly $\bar{w}$ (the $\sqrt{1 + 1/16}$ cancellation). The aim is exact by definition; all residuals are carried by the perturbation budget of the geometry contract below.

Lattice. Sites (n, k), $n + k$ even, $k \ge 0$, of a two-dimensional oriented lattice with children $(n \pm 1, k + 1)$, are realized as spatial boxes of side $2a$, $a := \bar{u}_x T_g/32$, with box core of half-width $a/2$, y-tracks spaced $\ell_y := 8a$ apart and a single z-track ($z \equiv 0$); epoch times are $t_k := kT_g$. Anchoring is *hybrid*: y, z on the global lattice, x relative to the parent (this absorbs drift), targets rounded to $\Lambda = (a/8)\mathbb{Z}^3$ (component error at most $a/16$).

Cones and cover. $D_\pm := \{w : \ w_x/|w| \ge 0.8, \ w_y/w_x \in \pm[1/8, 1/2]\}$ (the z component is constrained by the landing window); all parent velocities occurring in the construction lie in the *cover box* of half-width $0.05\,\bar{w}$ around $(\bar{u}_x, \pm\bar{u}_x/4, 0)$ — that the accepted candidates' velocity windows are contained in the cover is part of the machine check.

Landing window. A candidate entering the parent's radius-1 capsule at time $t_k + s$, $s \in [0, T_a]$, reaches the target core at t_{k+1} if and only if its velocity lies in $w_c \oplus [-h_0, h_0]^3$, where w_c is the required velocity of the perturbed instance and $h_0 := \frac{a/2}{T_g - s}$.

Site event (open bit). $G_{n,k}$: at time t_k the box core of site (n, k) contains at least N_h *qualified* particles (marked, alive, velocity in the cover box) *delivered by the tracked relay* — qualified particles of other provenance are never queried (measurability) — at least one of which has residual lifetime $\ge T_g$ (the *coverage clause*; by memorylessness each qualified particle satisfies this with probability $\beta := e^{-T_g/T_s}$, and the clause keeps the marked living set non-empty at every time between epochs). The relay from a good parent P^* (a coverage particle: residual lifetime $\ge T_g \supset$ entry window) reads entries into its radius-1 capsule during the entry window $[t_k, t_k + T_a]$ only.

C.3.2 The geometry contract

Lemma 2 (geometry contract; machine-certified)**.** *For $K_{\mathrm{mix}} \ge 20$ and all perturbations — parent position in core plus rounding, parent velocity in the cover box, entry offset $\le R$, window sweep $s \in [0, T_a]$, target rounding, branch signs — the landing window retains the margin*

$$h_0 - \mathrm{dev} - I_h - \mathrm{lip} \ \ge \ s^*,$$

where dev *is a rigorous upper bound on the deviation of $|w_c|$ from $\bar{w}$ and* lip *is the Lipschitz penalty of the s-grid cell; moreover the window box is contained in the cone $D_\pm$ and in the parent cover box. The certified slack is $s^* = 0.0048\,\bar{w}$ at $\tau = 0$ (worst case) and larger for $\tau > 0$ (Appendix A.3).*

Proof. Machine certification by interval arithmetic (verifier version 6; design in Appendix A): each perturbation is propagated as component intervals; Euclidean norms are bounded by the exact closed forms $\min \|x\|^2 = \sum_i \mathrm{dist}(0, [l_i, h_i])^2$ and $\max \|x\|^2 = \sum_i \max(l_i^2, h_i^2)$; the window sweep uses a 4096-cell grid with the Lipschitz penalty $|\partial_s w_{\mathrm{req}}| \le (\sup|v_p| + \sup|w_{\mathrm{req}}|)/(T_g - T_a)$ and the conservative left endpoint for h_0 (h_0 is increasing in s). □

C.3.3 The erosion patch

Lemma 3 (erosion patch)**.** *Set $\theta^* := s^*/(1.05\,\bar{w})$ and let $U := \{u \in S^2 : \angle(u, w_c/|w_c|) \le \theta^*\}$. Then the full patch $\{r\,u : u \in U, r \in I^*\}$ is contained in the landing window, and by isotropy*

$$\hat{\nu}(\text{accepted}) \;\ge\; \frac{1}{200} \cdot \frac{1 - \cos\theta^*}{2},$$

where $\hat{\nu} := \nu/q$ is the normalized shell measure.

Proof. For $u \in U$ and $r \in I^*$, using $|w_c| \le 1.05\,\bar{w}$ (certified) and chord $\le$ angle,

$$|r\,u - w_c|_\infty \;\le\; |r\,u - w_c| \;\le\; \big|r - |w_c|\big| + |w_c|\,\big|u - w_c/|w_c|\big| \;\le\; (I_h + \mathrm{dev}) + 1.05\,\bar{w}\,\theta^* \;\le\; h_0$$

by Lemma 2 ($\ell_\infty \le \ell_2$), so the patch lands in the window. The isotropic decomposition of $\hat{\nu}$ into radial mass (at least $1/200$ on I^*) times the uniform angular measure (cap mass $(1 - \cos\theta^*)/2$) gives the bound. That the patch lies inside $D_\pm$ is part of the interval check of Lemma 2. □

C.3.4 The disclosure capsule

Lemma 4 (disclosure-capsule identification)**.** *The construction reads, of the underlying randomness, only: the window-entry capsules $\mathcal{C} = \{(x_0, v) : \inf T_{P^*,\cdot} \in [t_k, t_k + T_a]\}$ of its sites, the pair coins of window entrants, the tracking judgments (indicator paths $(\mathbf{1}\{L > s\})$ only, truncated at $M = \lceil 2\mu \rceil$ tracked candidates, with μ as in Appendix C.3.7), and the private coins.*

Proof. (i) Under ballistic motion the contact set of an ordered pair is a single interval, so a window entry is a *first* contact and the pair (P^*, j) is unconsumed at that instant; pair-once is an absolute convention (re-marking never re-opens a pair), so the attempt of a window entrant fires inside the window with certainty and the construction reads only its outcome. (ii) Entries outside windows never appear in the construction's σ-algebra; the dynamics may consume other ordered pairs there and change mark states, but this does not affect pair-freshness at a new site, and unknown mark states are absorbed by the case analysis of Lemma 6 (interference is rescue-only). (iii) The window is the uniform T_a for every generation, the seed's site included. □

Consequence. The deduction tubes of Lemma 5 have segment length exactly $|v - v_P| T_a$: this is *disclosure geometry*, not dynamical exposure. The $\mathrm{Exp}(T_s)$-tail exposure extension caused by re-marking arises only on the dynamical side, which is the jurisdiction of Lemma 6, not of the disclosure.

Candidates. The candidates of a site are the window entrants whose pair (P^*, j) is *fresh*. Freshness of the randomness is guaranteed not by the definition but by the deduction: the window entrants of past sites — precisely the objects the construction has read — coincide by definition with the bad set of Lemma 5 and are removed by it exactly; the identity of the read σ-algebra with the deduction target is the crux of the bridge. No explicit exclusion of returning former parents or past entrants is needed.

C.3.5 The normalization bridge

Lemma 5 (normalization bridge; machine-certified)**.** *Candidate velocities carry the intensity* $\mathrm{d}\Phi(v) \propto \rho\,\pi R^2\,|v - v_P|\,\nu(\,\mathrm{d}v)$*; the relative-speed factor is bounded above and below by design constants and is accounted in the flux count of Appendix C.3.7. Measuring acceptance and deduction in the same normalized shell measure* $\hat\nu = \nu/q$*, with the bad set measured unconditionally on dynamics and mark states (geometrically,* v *restricted to the window box of its child branch; the eight-branch premise: the bad set is certified separately for each of the* $2^3 = 8$ *branch-sign combinations, namely the candidate's child branch and the* $\pm$ *branches of the two parents* P *and* P'*, taking the worst case, as in the certifier of Appendix A), the certified configuration satisfies*

$$\bar\Lambda\,\hat\nu(\mathrm{bad}) \;\le\; \tfrac{1}{2}\,\hat\nu(\mathrm{accepted}) \qquad \textit{for all } \tau \ge 0.$$

Proof. Deduction set. A disclosure intersection with a past site at generation gap $j \ge 1$ requires

$$D(v) := \tilde\Delta - v\,jT_g \;\in\; \mathcal{K}(v) := B(0,2R) \oplus \mathrm{seg}\big((v - v_P)T_a\big) \oplus \mathrm{seg}\big(-(v - v_{P'})T_a\big),$$

where $\tilde\Delta$ is the rounded site offset; D is affine in v with Jacobian $(jT_g)^3$, and drift is a translation, hence measure-invariant. Same-generation siblings ($j = 0$) are y-separated by $8a \gg \mathrm{diam}\,\mathcal{K}$ and deterministically do not intersect. The seed's capsule is counted as an ordinary site.

Box-hull Steiner upper bound. $W := (V \ominus P)\,T_a$ is a box (V the velocity window box, P the parent cover box, $\ominus$ the componentwise interval difference). For $x \in W$ the segment $\mathrm{seg}[0,x]$ lies in $\mathrm{conv}(\{0\} \cup W)$, which is contained in the component interval hull H (convexity, one line); applying this to each of the two segment factors gives hulls H_1, H_2, whence

$$\bigcup_v \mathcal{K}(v) \;\subseteq\; B(0,2R) \oplus H_1 \oplus (-H_2),$$

a box (edge lengths c_i = sums of the component edges) $\oplus$ ball of radius $r = 2R$, whose volume is the exact Steiner closed form

$$\mathrm{vol} \;=\; \prod_i c_i \;+\; 2r\sum_{i<j} c_i c_j \;+\; \pi r^2 \sum_i c_i \;+\; \tfrac{4\pi}{3} r^3$$

(the middle term carries the twelve edges with exterior dihedral angle $\pi/2$), with $\mathrm{diam} \le 4R + (\sum_i c_i^2)^{1/2}$. Ignoring the correlation of the common v is the superset covering $\bigcup_v [A(v) \oplus B(v)] \subseteq (\bigcup_v A(v)) \oplus (\bigcup_v B(v))$, hence conservative; the correctness of the box hull is the direct inclusion above (spot-reviewed).

Site count. An intersection at gap j requires $\tilde\Delta \in jT_g \cdot (\text{velocity window}) \oplus \mathcal{K}$, a position window of diameter approximately $j\,a\sqrt{3} + \mathrm{diam}\,\mathcal{K}$; with y-track spacing $8a$, a single z-track, and the gap j determining the generation (hence the x-multiplicity) uniquely, the multiplicity is in y only:

$$n_j \;\le\; \frac{\sqrt{3}}{8}\,j + 1 + \frac{\operatorname{diam}\mathcal{K}}{8a}, \qquad \sum_{j\ge 1} \frac{n_j}{j^3} \;=\; \frac{\sqrt{3}}{8}\,\zeta(2) + \Big(1 + \frac{\operatorname{diam}\mathcal{K}}{8a}\Big)\zeta(3) \;\approx\; 1.57.$$

The certified inequality. Combining,

$$\hat{\nu}\Big(\bigcup \text{bad}\Big) \;\le\; \bar{\Lambda}\cdot \frac{\operatorname{vol}(\mathcal{K})\,\sum_j n_j/j^3}{T_g^3\,\operatorname{vol}(S)}, \qquad \operatorname{vol}(S) = \frac{28\pi}{3}\,w_1^3,$$

with the shell endpoint $\bar{w} \in [w_1, 2w_1]$ taken at its worst separately on each side (deduction numerator at $\bar{w} = 2w_1$, geometric slack at $\bar{w} = w_1$). The certification gives $\bar{\Lambda}\,\hat{\nu}(\text{bad}) \le \frac{1}{2}\hat{\nu}(\text{accepted})$ at $K_0 = 20$, $T_a = T_s/32$ for all τ: at $\tau = 0$ (the worst case) the bridge closes up to $\bar{\Lambda}_{\max} = 7.8$, about twice the assumed bound, and intermediate τ are covered by monotonicity — the deduction is nonincreasing in τ (T_g^3 grows, no other term increases) and the acceptance is nondecreasing (the slack grows). The certified values are in Appendix A.3; the spot review reproduced the bridge numbers independently from the skeleton ($\operatorname{vol}(\mathcal{K}) = 66.5\,R^3$, $\sum_j n_j/j^3 = 1.569$, accepted mass 2.645×10^{-8}, $\bar{\Lambda}_{\max} = 7.79$; Appendix B). $\square$

C.3.6 Lemma X: private-coin coupling

Lemma 6 (private-coin coupling (Lemma X))**.** *Let $\mathcal{G}$ be the construction's entire disclosure (capsule point process, pair coins, tracking judgments). For a pair-fresh candidate j with a successful arrival at time α in epoch k,*

$$\mathbb{P}\big(j \textit{ is marked and alive at } t_{k+1} \;\big|\; \mathcal{G},\ \textit{arrival}\big) \;\ge\; e^{-(t_{k+1}-\alpha)/T_s} \;\ge\; \beta.$$

Proof (three observations; review cycles 5–8). *Observation 1 (failure requires a private coin).* Write $\bar{Q}_j$ for the disqualification of j at t_{k+1}. At arrival, j is in one of three states, and in each of them $\bar{Q}_j$ implies the shortfall of a single private exponential coin of j: (i) unmarked — the arrival marks it, and disqualification requires the fresh lifetime $L_j < t_{k+1} - \alpha$; (ii) marked and dead — *re-marking is allowed*, the arrival re-marks it, as in (i); (iii) alive — the arrival is a no-op but the qualification is already held, and disqualification requires the current residual $R_j < t_{k+1} - \alpha$. In all cases $\bar{Q}_j \subseteq A_j :=$ {the private coin of j falls short of $t_{k+1} - \alpha$}. *Observation 2 (joint independence).* All conditioning up to time α reads lifetimes only through indicator paths $(\mathbf{1}\{L > s\})_{s\le\alpha}$; the exponential law is memoryless under $\sigma(\min(L, s_0))$-type conditioning, so the fresh coins $\{L_j\}$ and the residuals $\{R_j\}$ remain jointly independent $\operatorname{Exp}(T_s)$ conditionally on $\mathcal{G}$, and $\mathbb{P}(A_j \mid \mathcal{G}) \le 1 - e^{-(t_{k+1}-\alpha)/T_s}$. *Observation 3 (post-arrival interference is rescue-only).* Rogue activity after α produces only additional arrivals at j; an additional arrival is a no-op if j is alive and a re-mark if j is dead — there is no path that creates a disqualification. Hence the inclusion of Observation 1 survives all later interference. $\square$

Remark 2. *No monotone comparison of whole processes (thinning, censored comparison) is used anywhere; the additivity of marks — a mark never kills and never displaces — is what makes failure one-directional. This is how the construction escapes the non-monotonicity trio recorded in Appendix B.*

C.3.7 Same-branch flux

One-generation relay. From the good parent P^*, the relay delivers to the target core the window entrants that are pair-fresh (Appendix C.3.4), have velocity in the accepted patch $\times I^*$ band (Lemma 3), succeed their pair coin (probability p), and remain marked and alive from arrival (delay τ) to t_{k+1} (Lemma 6); landing in the core is then the content of the window definition (Appendix C.3.1).

Flux lower bound, same-branch correction included. Children of the opposite branch satisfy $|v - v_P| \geq 0.4\,\bar{w}$. For children of the same branch, remove from the patch the ball of radius $\theta^*\bar{w}/2$ centred at v_P: at least a 1/4 solid-angle fraction of the patch survives, and on it $|v - v_P| \geq \theta^*\bar{w}/2 > 0$. In both cases the coefficients are functions of the design constants at fixed K_0 only — not of m_1 or ρ (no circularity; cycle-11 audit). Hence the number of candidates stochastically dominates a Poisson variable of mean

$$\mu \;\geq\; c_4(\tau/T_s)\,\frac{m_1}{32},$$

where $c_4(\tau/T_s) > 0$ is a constant of the certified design (the 1/32 is $T_a = T_s/32$), and by Lemma 5 the pair-fresh candidates surviving the deduction retain at least the mean $\mu/2$.

Open bit. The site is open when the tracked relay has delivered at least

$$N_h \;:=\; \lceil \beta\mu/16 \rceil$$

qualified particles (Appendix C.3.1); the tracking truncates observation at $M = \lceil 2\mu \rceil$ candidates. By Lemma 6 the failures satisfy $\sum_j \mathbf{1}_{\bar{Q}_j} \leq \sum_j \mathbf{1}_{A_j}$, a sum stochastically dominated by independent Bernoulli$(1-\beta)$ variables; Chernoff bounds give

$$\mathbb{P}(\text{site open} \mid \mathcal{G}) \;\geq\; 1 - e^{-c\mu} \;=:\; 1 - \varepsilon(m_1) \;\longrightarrow\; 1 \qquad (m_1 \to \infty)$$

— the generous slack of $N_h = \lceil \beta\mu/16 \rceil$ against the surviving mean $\geq \beta\mu/2$ also absorbs the truncation overflow $\mathbb{P}(\text{Poisson}(\mu) > 2\mu) \leq e^{-c'\mu}$.

C.3.8 Assembly

By Lemma 6 and the flux bound, $\mathbb{P}(\text{open} \mid \mathcal{G}) \geq 1 - \varepsilon(m_1)$ holds *conditionally on the entire disclosure*; by the tower property the same lower bound holds sequentially in any breadth-first order of the sites, so the open bits stochastically dominate an i.i.d. Bernoulli$(1-\varepsilon)$ field on the oriented lattice (direct sequential conditioning; the domination theory of [13] is not needed). Since $\varepsilon(m_1) \to 0$, for $m_1 > C_0$ we have $1 - \varepsilon > 3/4$.

Identification of the comparison lattice. The construction opens sites of the oriented graph $\mathcal{L} = \{(n,k) \in \mathbb{Z} \times \mathbb{Z}_{\geq 0} :\ n+k \text{ even}\}$ with oriented edges $(n,k) \to (n\pm 1, k+1)$; openness is a property of the site alone, and each site has exactly the two children $(n \pm 1, k+1)$ and the two parents $(n \mp 1, k-1)$. Liggett's bound [12] is proved for oriented site percolation on $\mathbb{Z}^2$: sites open independently with probability α, percolation meaning an infinite oriented path from the origin moving in the positive x and y directions; his Theorem 1(b), applied with $q = p = \alpha$ through his exact identification of the growth chain with the diagonals $x + y = n$, gives percolation for every $\alpha \geq 3/4$, i.e. $\alpha_c \leq 3/4$. The two graphs are identified by the bijection $\varphi(x,y) := (x-y,\, x+y)$, which maps $\mathbb{Z}^2_{\geq 0}$ onto the cone $\{(n,k) \in \mathcal{L} : |n| \leq k\}$, carries the step

$(x, y) \to (x+1, y)$ to $(n, k) \to (n+1, k+1)$, the step $(x, y) \to (x, y+1)$ to $(n, k) \to (n-1, k+1)$, and the origin to the seed site $(0, 0)$. Every oriented path in $\mathcal{L}$ from $(0, 0)$ stays in this cone ($|n|$ changes by one per step while k increases by one), so reachability from the seed site is decided inside the cone; φ is an isomorphism of oriented graphs onto exactly this cone, and it carries i.i.d. Bernoulli site openness to i.i.d. Bernoulli site openness. Hence the bound transfers verbatim: site openness probability $> 3/4$ implies an infinite open oriented cluster of the seed site with positive probability.

Consequently, for $m_1 > C_0$ the open bits percolate. Through the coverage clause each open site keeps a marked living particle at every time of its epoch, so along an infinite open path the marked living set is non-empty at *every* time — stronger than the definition of survival (non-emptiness at arbitrarily large times, Section 1) — and the full model survives. *Ignition*: from the arbitrary-velocity seed, a bootstrap of a few generations assembles, with positive probability, the first N_h-herd in a first site; survival is a positive-probability statement, so this suffices.

Uniformity of the constants. No upper bound on m_1 is used anywhere: the coverage clause, the Chernoff step and the observation truncation $M = \lceil 2\mu \rceil$ improve or are unaffected as μ grows; the cost terms are nonincreasing in $(K_{\text{mix}}, \tau/T_s)$ term by term. C_0 is assembled explicitly from $\{c_4, \beta, \text{the bridge } 1/2, \text{the Chernoff constant}\}$; its explicit quantification is deliberately left open (Section 10). Intermediate τ are covered by the monotonicity of both sides of the bridge (Lemma 5). □

Consequences. With Theorem A the existence of the critical surface (layer 1) is settled from both sides; the certified refutation of the SIR-type bound (Section 4) sharpens the intermediate band, while the separation of the thresholds themselves remains open (Section 10). With Theorem B and the scale family of Section 7, Theorem C2 yields Theorem D (Appendix C.5).

C.4 Theorems C6a and C6b′: fronts

Throughout, the front radius $R(t)$ is the supremum of $|x_j(t)|$ over the particles j ever marked by time t (dead marks included).

Theorem C6a (superlinear front). Assume $T_s = \infty$, $p > 0$, and that ν is isotropic with unbounded speed support. Fix $W > 0$ with $\rho_W := \rho\, \nu(\text{shell}_W) > 0$, where $\text{shell}_W := \{w : |w| \in [W, W+1]\}$; unbounded support supplies arbitrarily large such W. *Condition on the seed's velocity* v_0 — unconditionally the counts below are mixed Poisson; conditioning removes the mixture. The seed is marked forever ($T_s = \infty$), and class-W particles enter its radius-R capsule at rate

$$\lambda_W(v_0) \;=\; \rho_W\, \pi R^2\, \mathbb{E}_{w \sim \nu_W} |v_0 - w|,$$

ν_W being the normalized restriction of ν to shell_W. Positivity $\lambda_W(v_0) > 0$ holds for ν-a.e. v_0; this is the point where isotropy is consumed: a rotation-invariant law has no atoms away from the origin, so $\nu_W(\{v_0\}) = 0$ and $\mathbb{E}_{w \sim \nu_W} |v_0 - w| > 0$ (for atomic, non-isotropic ν this step can fail). *Encounters are attempts*, on three grounds: the seed is immortal, so every contact window meets its alive interval and every entry triggers an attempt; motion is ballistic, so each ordered pair meets in a single contact interval and every entering pair is fresh, attempted once and only once; and each ordered pair carries an independent p-coin. The entries are the points of the phase-space Poisson process falling in the capsule swept along the seed's trajectory, so the number of *successful* transmissions to class-W particles by time t is Poisson of mean

$p\,\lambda_W(v_0)\,t \to \infty$: almost surely some class-W particle j is marked, at some finite time t_W. For $t \ge t_W$,

$$|x_j(t)| \;\ge\; |v_j|\,(t - t_W) - |x_j(t_W)| \;\ge\; W\,t - \big(W\,t_W + |x_j(t_W)|\big),$$

and the deficit is a *per-particle constant*: no uniformity over particles is needed. Since marks never die, $R(t) \ge |x_j(t)|$, so $\liminf_{t\to\infty} R(t)/t \ge W$ almost surely; intersecting over a countable sequence $W \uparrow \infty$ gives $R(t)/t \to \infty$ almost surely. □ Whether the isotropy hypothesis can be removed is unknown.

Theorem C6b′ (linear bound for bounded speeds). Assume $|v| \le w_{\max}$ a.s. and $\tau > 0$; let x_0 be the seed position. *Well-foundedness.* Every mark event other than the seed has a unique parent mark event (a re-mark is an ordinary transmission, delay τ included), and its attempt time satisfies $a = \max(\alpha, m_{\text{parent}}) \ge m_{\text{parent}}$, so marking times increase by at least τ along any ancestry: $m_{\text{child}} = a + \tau \ge m_{\text{parent}} + \tau$. This simultaneously proves that ancestries are *well-founded* — no cycles, no infinite descent — which, and not locality per se, is the true role of $\tau > 0$. A mark event at time $m_n \le t$ therefore terminates a finite ancestry $j_0 = \text{seed}, j_1, \ldots, j_n$ with attempt times $a_1 < \cdots < a_n \le t$ and $n\tau \le m_n \le t$, i.e. $n \le \lfloor t/\tau \rfloor$. *Telescoping travel accounting.* Each attempt happens inside the contact window, so parent and child are within distance R at the attempt time: $|x_{j_i}(a_i)| \le |x_{j_{i-1}}(a_i)| + R$ for $1 \le i \le n$; and between consecutive charging times each particle moves at most at speed $w_{\max}$: $|x_{j_{i-1}}(a_i)| \le |x_{j_{i-1}}(a_{i-1})| + w_{\max}(a_i - a_{i-1})$. Iterating down the ancestry from $|x_{j_n}(t)| \le |x_{j_n}(a_n)| + w_{\max}(t - a_n)$ to the seed ($|x_{j_0}(a_1)| \le |x_0| + w_{\max}\,a_1$), the travel time telescopes exactly across the partition $[0, a_1] \cup [a_1, a_2] \cup \cdots \cup [a_n, t]$ of $[0, t]$ — each particle of the ancestry is charged once, the child's travel between attempt and mark arrival included, with no double counting — while each of the n hops contributes a spatial jump at most R:

$$|x_{j_n}(t)| \;\le\; |x_0| + w_{\max}\,t + nR.$$

Substituting $n \le \lfloor t/\tau \rfloor$ and taking the supremum over mark events with $m_n \le t$,

$$R(t) \;\le\; w_{\max}\,t + R\left\lfloor \frac{t}{\tau} \right\rfloor + |x_0|.$$

The bound uses only the speed cap after marking, so it applies to dead-marked particles as well. As a byproduct, the front radius is finite — deterministically so — at every time. □

The $\tau = 0$ pathology. At $\tau = 0$ the bound degenerates, and genuinely so: if the open cluster of the seed in the p-bond-diluted instantaneous geometric graph at time 0 is infinite (for fixed $p > 0$ this has positive probability at ρR^3 large, by standard renormalization), the whole cluster is marked at time 0^+ and $R(0^+) = \infty$ can occur. $\tau > 0$ is precisely what guarantees locality of the front in dense regimes — physically, the probe travel time.

C.5 Theorem D: motion beats static geometry

Normalize $R = 1$ (Section 2; all hypotheses below are stated in this normalization) and fix any ρ with $0 < \rho < \rho_c^{\text{RGG}}(p{=}1)$. *Static side.* By Theorem B the survival event coincides with percolation of the seed's cluster in $G(\rho, 1; p)$; the standard monotone coupling in p and subcriticality at $p = 1$ give almost-sure extinction for every $p \le 1$, T_s, τ. *Moving side.* Fix any $p > 0$, $\tau \ge 0$, $T_s > 0$ and a base law ν^0 satisfying the hypotheses of Theorem C2 (isotropy, shell mass $q > 0$, $\bar\Lambda \le 4$), and let $\nu_s := \nu^0(\cdot/s)$ be its pushforward under $w \mapsto sw$.

Lemma 7 (scale invariance)**.** *ν_s has shell $S_s = sS^0$ and shell mass $q_s = \nu^0(S^0) = q$ (invariant); its density bound scales as s^{-3} and the shell volume as s^3, so the relative density ratio $\bar{\Lambda}_s = \bar{\Lambda}^0$ is invariant. Both q and $\bar{\Lambda}$ are uniform in w_1; τ/T_s is untouched, so $C_0(\tau/T_s)$ is fixed, while $K_{\mathrm{mix}} = s\,w_1^0 T_s$ and $m_1 = p\rho q\pi\, s\,w_1^0 T_s$ grow linearly in s.*

Proof. The pushforward under $w \mapsto sw$ scales lengths by s, hence $S_s = sS^0$ and $\mathrm{vol}(S_s) = s^3 \mathrm{vol}(S^0)$; masses are preserved, so $q_s = q$ and $\sup \mathrm{d}\nu_s/\,\mathrm{dLeb} = s^{-3} \sup \mathrm{d}\nu^0/\,\mathrm{dLeb}$; the ratio $\bar{\Lambda} = (\text{density bound})/(q/\mathrm{vol}(S))$ is therefore invariant. The remaining statements are read off the definitions of K_{mix} and m_1. □

For s large enough, $K_{\mathrm{mix}} \geq 20$ and $m_1 > C_0(\tau/T_s)$, so by Theorem C2 the full model survives with positive probability. Under the entry-driven convention, sufficiently fast motion enables colonization at densities where static geometry forbids it for every p. □

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